

OpperTune: Post- deployment config tuning

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Parameter tuning

- Challenges:
 - Determining the range of values
 - Knobs interact with each other
 - Noisy measurements
 - Dynamic workloads
 - Dependent on hardware and environment

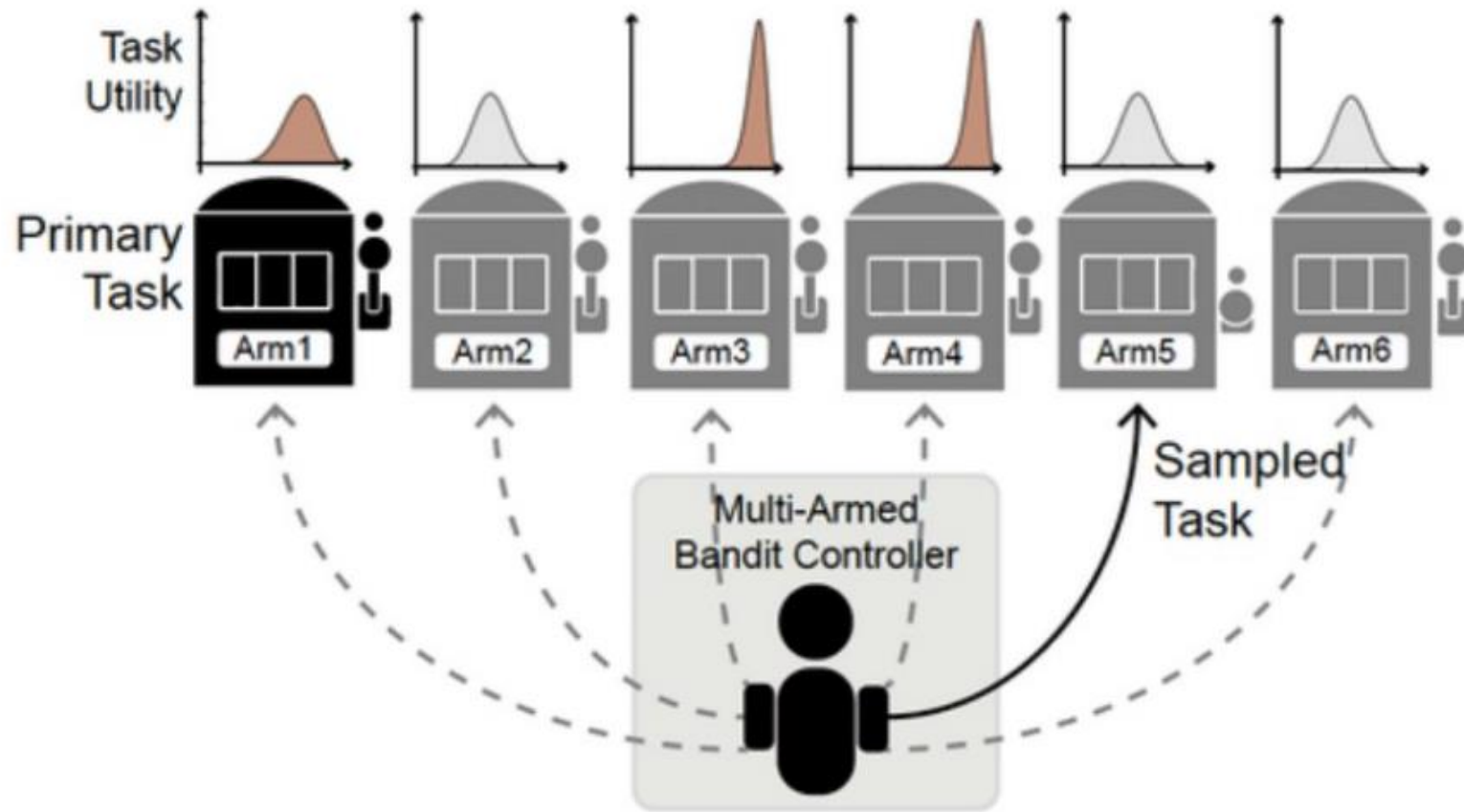
Autotuning

- Manual heuristics
- Offline profiling or grid search
- Bayesian optimization
- Bandit-based methods
- RL-based methods

Online tuning

- Offline tuning: benchmarking, profiling, BO etc.
- Online tuning: adapts during runtime
- Need to be careful to not affect application
- Challenges:
 - Delayed feedback?
 - Interference from co-located workloads
 - Stability concerns
 - Limited budget

Bandit algorithms



Discussion

1. If you could combine offline tuning with online tuning, how would you do it?